julia racer — iRacing test plan

IRPLAN-2, 17 October 2026 — everything julia still needs from iRacing's Lotus 49

How to use this plan

The plan is in six parts, about **2 hours** in total. Any part can be done on its own, in any order. Parts A–C are in the garage (no driving); D–F are on track.

Part	What	Time	What it settles in julia
A	Export your 32 saved setups as setup sheets	15 min	Every knob's exact value and limits
B	Garage ranges not yet captured	20 min	The † (provisional) knobs in julia's Garage tab
C	Garage couplings	15 min	Does camber, toe or pressure move the corner weights? How stiff is each bar?
D	The physics tests from 15 October (skidpad, pulls, coasts)	30 min	The physics item (PHYSFID-1)
E	New on-track tests: aero, rear traction, roll	25 min	Nose lift, the 35° diff, the bar split
F	Other cars: contact and drafting	20 min	CARINTERACT-1

Car: Lotus 49, **track:** Monza GP *without* chicanes (as on 10–17), unless a test says Skidpad. Base setup: your 261010_rung_official.

Before you start (2 minutes)

1. **Telemetry on** (Alt+L) for every on-track test. One file per test is ideal: stop and restart telemetry between tests.
2. **Screenshots in the garage — wait 3 seconds after each click.** On 10–17, two screenshots showed iRacing's *previous* solve, not the one for the knob shown (the LR 30 N/mm shot is really about 40 N/mm). Wait until the numbers stop changing and the labels lose their asterisk (Spring rate:*), then take the shot.
3. **One knob at a time, from the base setup.** Reload 261010_rung_official before each new knob.
4. **Filing:** put everything on the USB stick under 2610xx/telemetry, 2610xx/screenshots, 2610xx/setups, 2610xx/sheets. Write a short note of anything you noticed (a text file is fine).

Part A — Setup sheets (15 min, no driving)

The .sto files iRacing saves are encrypted, so julia cannot read the 32 setups you saved on 10–17. A setup **sheet** lists every value in plain text.

1. In the garage, load each of the 32 setups (261010_rung_official_<knob>_<end>), then export it as a sheet: the **Export** button, or the save dialog's text/HTML copy option if it offers one.
2. Do the same for 261010_rung_official itself and for 261005_jr_ww103.
3. Copy the exported files to 2610xx/sheets.

If iRacing offers no sheet export, a screenshot of each tab (Tires, Chassis, Drivetrain) per setup does the same job.

Part B — Garage ranges (20 min, no driving)

For each knob: click to its **minimum**, wait, screenshot; then to its **maximum**, wait, screenshot; then reload the base. For drop-down knobs, open the list and screenshot it (that shows every option).

#	Knob	Where	How
B1	Cold pressure (one corner)	Tires	min / max
B2	Camber (LF, then	Chassis, corners	min / max

#	Knob	Where	How
	LR)		
B3	Toe-in front, toe-in rear	Chassis, Front / Rear	min / max
B4	Brake bias	Chassis, Front	min / max
B5	Steering ratio	Chassis, Front	open the list (or min / max)
B6	Rear ARB diameter	Chassis, Rear	open the list
B7	Final drive	Drivetrain	open the list
B8	First … Fifth gear	Drivetrain	open each list
B9	Clutch plates, preload, drive ramp, coast ramp	Drivetrain	min / max each

Part C — Garage couplings (15 min, no driving)

Screenshot the **Chassis** tab with all four corners in view after each step (wait 3 s each time).

#	Do	What it shows
C1	LF camber: +5 clicks, then −5 clicks (from base)	Does a camber change move the corner weights or ride heights? (Two of your old setups suggest it moves weight sideways.)
C2	Front toe: +3 clicks, then −3	The same for toe
C3	LF cold pressure: minimum, then maximum (Tires tab, then Chassis tab)	How tyre pressure moves the corner weights
C4	**Big front twist:** LF perch +25 clicks (about +10 mm). Then, with that twist kept, step the front	The front bar's stiffness per hole and per diameter. On 10–17 the twist was only 2 mm, too

#	Do	What it shows
	ARB arms through soft, 2, 3, 4, 5, 6, firm, then the diameters (Disconnected, 0.375, 0.4375)	small to read.
C5	**Big rear twist:** reload; LR perch +25 clicks; step the rear arms and diameters the same way	The rear bar

Part D — The 15 October physics tests (30 min)

Unchanged from iracing_tests_261015.txt:

- ☐ **D1 Skidpad limit, both setups** (10 min). Skidpad, default setup: drive the circle clockwise at a steady speed, raising the speed in small steps (hold each step about 5 s) until the front starts to run wide. Repeat anticlockwise. Repeat both with WW103.
- ☐ **D2 Full-throttle pulls in 4th, both setups** (10 min). Monza main straight: from about 120 km/h in 4th, full throttle until the braking point. Three pulls per setup.
- ☐ **D3 Coast in gear, both setups** (5 min). At about 250 km/h lift completely (clutch in gear, no brake) and coast to 100 km/h. Twice per setup.

Part E — New on-track tests (25 min)

- ☐ **E1 Steady speeds on the straight** (8 min, default setup). Monza main straight: hold **150, 200 and 250 km/h** each for 5 seconds with a steady throttle (as steady as you can), then lift completely at 250 and coast to 150. Two runs. Then repeat once with the front springs at **35 N/mm**. *Why:* on 10–17 the nose rose 4.7–14 mm at 250 km/h depending on the spring. Holding a steady speed separates the aerodynamic lift from the load the drag moves to the rear; julia's nose currently rises too much on stiff springs.
- ☐ **E2 Corner exits on power, both diff ramps** (10 min). Drive ramp **35°**: five exits each from Parabolica and Lesmo 2, a smooth throttle squeeze from the apex, no wheelspin on purpose. Then the same at **75°**. *Why:* at 35° iRacing's outside rear runs about 8 % ahead while the inside rear keeps its

grip; in julia the inside rear spins up. More exits than on 10–17 (6 at 35°) make it a fit, not a hint.

- ☐ **E3 Roll per bar on the skidpad** (7 min). Skidpad, default setup, a steady circle at about 0.8 g for 10 s, once each with: front bar **Disconnected**, **0.375 arms 2** (base), **0.4375 firm**; then the rear bar **Disconnected** and **firm** (front back to base). *Why:* the body roll per g at each bar setting tells how much of the car's roll stiffness each bar carries. julia currently judges that split (46 % front).

Part F — Other cars: contact and drafting (20 min)

As Part 2 of iracing_tests_261015.txt: an AI race (Lotus 49 if AI is offered, lowest AI skill, 3–5 cars, practice session, damage on). **F1 contact:** a light nudge at about 5 and 15 km/h closing speed, a side-by-side lean, and (optional, last) a 30 km/h hit; one slow lap after each, and a sentence on what you saw (pull, damage, incident count). **F2 drafting:** two lone runs down the main straight; two runs right behind an AI car, pulling out about 50 m before the braking point; then one run each at about 10, 20 and 40 m behind.

What julia does with the files

Files	Tool (JuliaMotorMTK/tools)	Result
A, B screenshots / sheets	garage knob table	the † ranges in the Garage tab become iRacing's
C screenshots	garage_fit_261017.jl (new frames)	camber/toe/pressure couplings; each bar's rate
D	irgold_steady.jl, longval, coast_compare.jl	front/rear grip split, drive and engine-braking losses
E1	aeroheave_cmp_261017.jl	lift vs drag transfer
E2	difframp_261017.jl, difframp_replay_261017.jl	the 35° diff and rear traction
E3	arbfit_261005.jl	the bar split

Files	Tool (JuliaMotorMTK/tools)	Result
F	irtest_contact_261015.jl, irtest_draft_261015.jl	contact law, draft shape